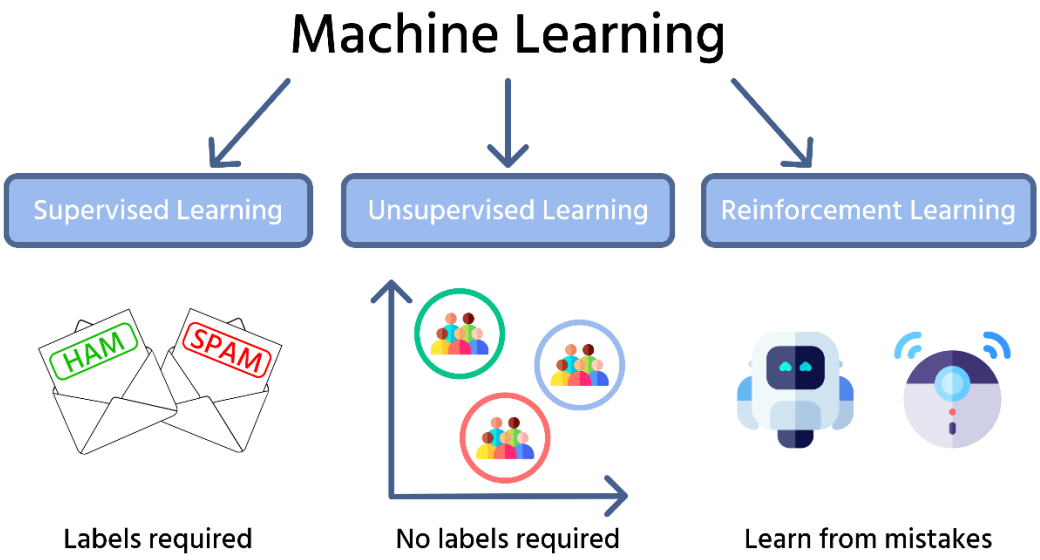
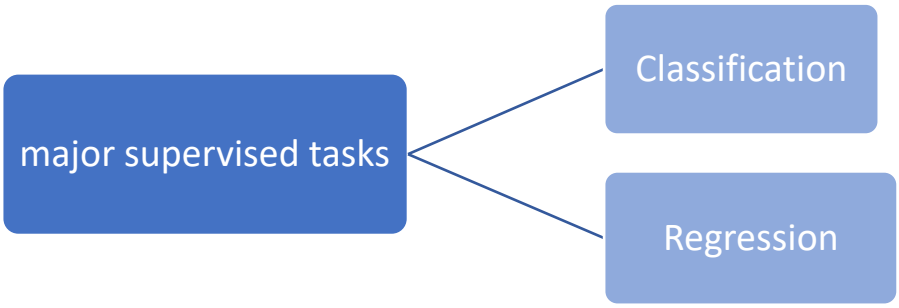
**Types of ML:**




**1)Supervised Learning**

The model is trained on a **labelled dataset**, meaning every input comes paired with the correct output. The system learns the underlying relationship between inputs and outputs to predict outcomes for completely new, unseen data.




**i)Classification:** Assigns data into discrete categories.

Examples:

- Email spam detection: Spam or Not Spam
- Sentiment analysis: Positive, Negative, or Neutral
- Image classification: Cat or Dog

Labelled dataset example:

| Email ID | Email Text | Spam (Label) |
|---|---|---|
| 1 | “Win a free iPhone now!!! Click here” | Spam |
| 2 | “Team meeting at 2 PM today” | Not Spam |
| 3 | “Limited time offer – claim your prize” | Spam |
| 4 | “Here is the report you requested” | Not Spam |
| 5 | “Congratulations! You’ve won $1,000” | Spam |

**ii)Regression:** Predicts continuous, numerical values (e.g forecasting house prices or stock trends).

Common algorithms:

- Linear Regression
- Decision Tree
- Random Forest

Example:

You can predict annual salary using years of experience.

| Years of Experience (x) | Annual Salary in Lakhs₹(y) |
|---|---|
| 1 | 4.5 |
| 2 | 6.0 |
| 3 | 7.5 |
| 4 | 9.0 |
| 5 | 10.5 |

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**2)Unsupervised learning**

The model works with unlabeled data. The system explores the data independently to uncover hidden structures, patterns, or anomalies.

**Techiques:**

**i)Clustering:** process of grouping data points into clusters based on their similarity.

Items within same cluster →highly similar

Items in different cluster → distinct

Algorithm: K-means clustering

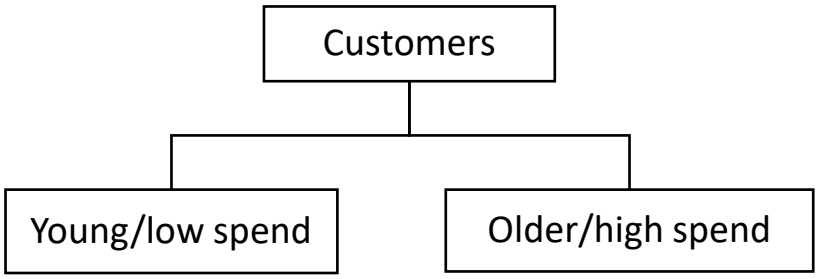
Customer A → age 21, spends ₹2,000

Customer B → age 23, spends ₹2,500

Customer C → age 55, spends ₹30,000

Customer D → age 52, spends ₹28,000

**Result: Customer segmentation**




**ii)Dimensionality Reduction Techniques**

Helps reduce the number of features while preserving important information.

Example: Principal Component Analysis (PCA).

**iii) Association Rule Learning**

Association rules discover co-occurrence, hidden relationships and "if-then" dependencies between variables in massive datasets.

Example:

Customers who purchase bread may also frequently purchase butter.

An association rule can be represented as:

Bread → Butter

Common algorithm: **Apriori**

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**3)Reinforcement learning**

trains an agent to make a sequence of decisions through trial and error. The agent interacts with the environment, receives feedback in the form of rewards or penalties and learns optimal actions over time. Over time, it learns a strategy called a policy for selecting actions that maximize cumulative reward.

**Key Components**

- **Agent:** The learner or decision-maker.
- **Environment:** The world in which the agent operates.
- **Action:** A decision made by the agent.
- **Reward:** Feedback received after an action.
- **Policy:** The strategy used to choose actions.
- **Policy:** The strategy used to choose actions.

**Example: Training a Game-Playing Agent**

A game-playing agent can try different moves.

- A successful move may earn a positive reward.
- An unsuccessful move may receive a negative reward.
- By repeatedly interacting with the game, the agent can learn a strategy that improves its performance.

Other applications include robotics, game AI, and certain resource-allocation problems.

GitHub Repository — AI Essentials